

(a)

carbonyl migration removed

Threose	1 → 1	1 → 1	3 → lost	3 → lost	—
Ribose	2 → 2	2 → 2	11 → lost	—	—
Glycerol	2 → 2	2 → 2	—	—	—
Deoxyribose	3 → lost	3 → 4	11 → lost	—	—
Glyoxylate	6 → lost	3 → 3	11 → lost	5 → lost	—
	F	FA	G	GA	PA

(b)

Cannizzaro reaction removed

Threose	1 → 1	1 → 1	3 → 3	3 → 3	—
Ribose	2 → 2	2 → 2	11 → 11	—	—
Glycerol	2 → lost	2 → lost	—	—	—
Deoxyribose	3 → 5	3 → 5	11 → 11	—	—
Glyoxylate	6 → 6	3 → lost	11 → 12	5 → lost	—
	F	FA	G	GA	PA

reached in the same number of steps still reached, but by a longer route no longer reachable not reachable with the whole rule set